INSTRUCTIONS

GEK-8081

MAGNET-OPERATED TWO-SHOE D-C BRAKE

IC9528A106

Before any adjustments, servicing, parts replacement or any other act is performed requiring physical contact with the electrical working components or wiring of this equipment, the POWER SUPPLY MUST BE DISCONNECTED.

GENERAL

The IC9528A106 brake is a spring-set power-released magnet brake intended for operation on the shaft of a d-c or a-c motor. A d-c magnet on the brake overcomes spring pressure to release the brake when energized.

Ratings of the IC9528A106 brake are shown in Table I.

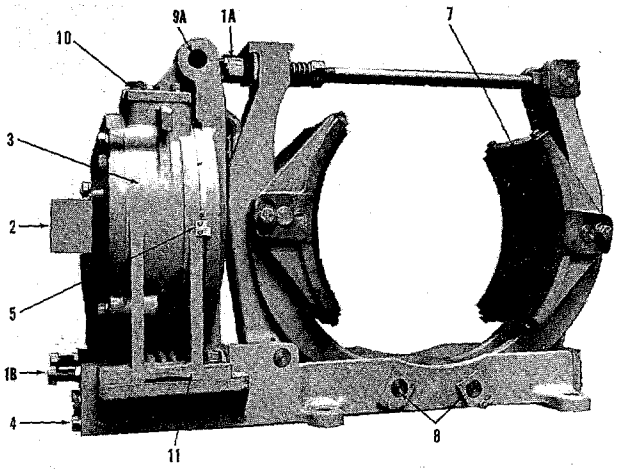


Fig. 1. IC9528A106 d-c brake

TABLE I RATINGS

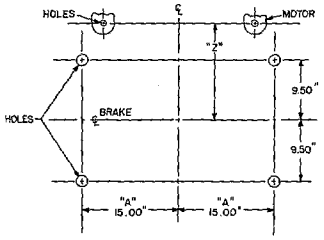
Wheel Diameter in Inches	Torque Rating in Lb-Ft					IC9528 Form
	Series-wound		Shunt-wound			
				Continuous		
	30-min	60-min	60-min	Without Forcing	With Forcing	
30	9000	6000	9000	6750	9000	A106

INSTALLATION

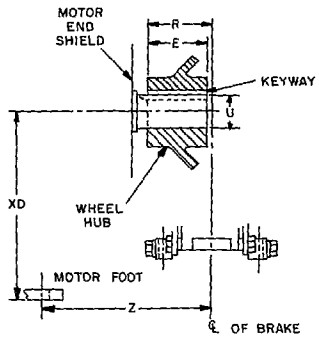
Because of its physical size and weight, this brake is designed primarily for floor mounting.

Locate the height of the brake-mounting bedplate for the brake approximately 20-3/4 inches from the centerline of the motor shaft, allowing an additional 1/8- to 1/4-inch for shimming purposes. When the brake is placed on the bedplate, the brake may tilt due to the weight of the magnet. Therefore, it is recommended that the mounting bolts be partially tightened before releasing lifting cables.

To locate mounting holes for the brake, refer to Fig. 2. Dimensions "A" in Fig. 2A must line up the brake mounting holes on each side parallel to the centerline of the motor.



A. Brake in relation to General Electric motor



B. Brake in relation to motor other than General Electric

Fig. 2. Relation of motor to brake for proper installation

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

Dimension "Z" can be obtained from Table II if the motor is of General Electric manufacture. If it is not a General Electric motor, dimension "Z" can be obtained by use of the motor outline (See Fig. 2B) together with the dimensions indicating the position of the brake wheel on the motor shaft in Table II.

TABLE II DIMENSIONS-BRAKE TO MOTOR

Pt. No.	Motor Frame	XD	Z	U	Keyway WD	Keyway DP	E	R	Drawing No.
1	MD620	20 7/8	18 3/4	5.875	1 1/2	3/4	6 3/4	10 1/4	139B8184P1
2	MD622	23	18 1/2	6.250	1 1/2	3/4	7 1/4	10 3/4	139B8184P2
3	MD624	24	19 1/4	7.000	1 1/2	3/4	9 1/4	10 3/4	139B8184P3

MOUNTING THE WHEEL

Assemble the brake wheel to the motor shaft before setting the brake in position. For straight-bore wheels, press the wheel on the shaft to the point where the recessed hub face lines up with the end of the motor shaft. For taper-bore wheels, place the wheel on the motor shaft and fasten in place with the nut and locking device furnished with the motor.

MOUNTING THE BRAKE

Set the brake on the bedplate. Back off on the gap adjustor "D" to spread the brake shoes. Adjust the height by means of shims so that a straight edge connecting the centerlines of the two shoe pivots passes through the centerline of the motor shaft. (See Fig. 3.)

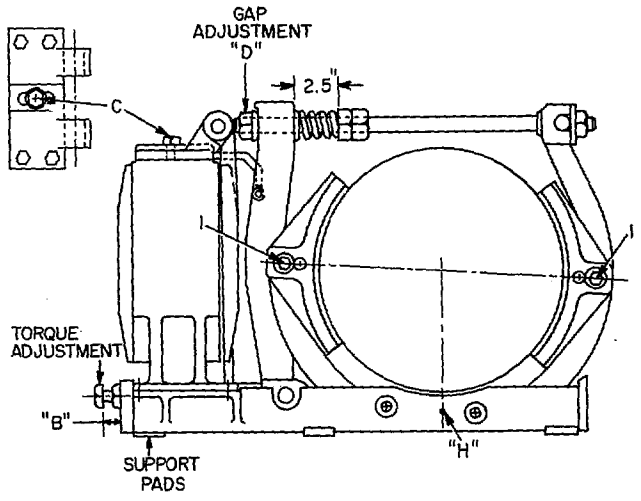


Fig. 3. Brake gap adjustment

Position the brake so that a plumb bob dropped from the centerline of the motor shaft is in line with the aligning hole "H" in the base of the brake.

Before tightening mounting bolts, loosen all four shoe bolts (1) (see Fig. 3). Also, loosen the equalizer bolt "C" at the center of the bracket.

TORQUE ADJUSTMENT

Set the two brake springs for the torque required by the application. In most instances, a torque approximately equal to the full-load torque of the motor will be satisfactory.

See Table III for torque settings and spring length. To set torque for a given value, turn both torque adjustment bolts in or out until the distance "B" from the back side of the bolt head to the end block is that indicated in Table III. Tighten the torque adjustment locknut and shoe bolts after the adjustment is made. If the full-load torque of a motor is not known, it can be determined from the following formula.

$$\text{Torque in lb-ft} = \frac{5250 \times \text{HP}}{\text{FL Speed in RPM}}$$

Braking torque as calculated by the above method is based upon linings being "worn-in" sufficiently to obtain at least 85-percent surface contact with the brake drum.

TABLE III TORQUE AND SPRING LENGTH

Torque (lb-ft)	Dim. "B"	Spr. Length (Ref)
9000	2.25	12.46
8000	2.75	12.65
7000	2.92	12.82
6750	2.96	12.85
6000	3.12	13.00

MAGNET GAP ADJUSTMENT

After mounting the brake, turn the gap adjusting nut "D" (see Fig. 3) until the edge of the armature "F" is in-line with the face of the notch "E" (Fig. 4). When the lining wears, the armature will move out to the outer end of the gage "G". The armature gap should not be greater than indicated by "G" or the brake may not release with low current.

EQUALIZATION OF SHOE CLEARANCE

With the torque set and all other bolts tightened, the equalizer bolt "C" Fig. 3 on center of the top of the magnet housing holding the "L" bracket should be tightened. The equalizer bolt "C" must be loosened each time the magnet gap is readjusted to maintain approximately equal lining to the wheel clearance; then, retighten it after readjustment is accomplished. Reset the tie-rod spring length to 2.50 inches. (See Fig. 3.)

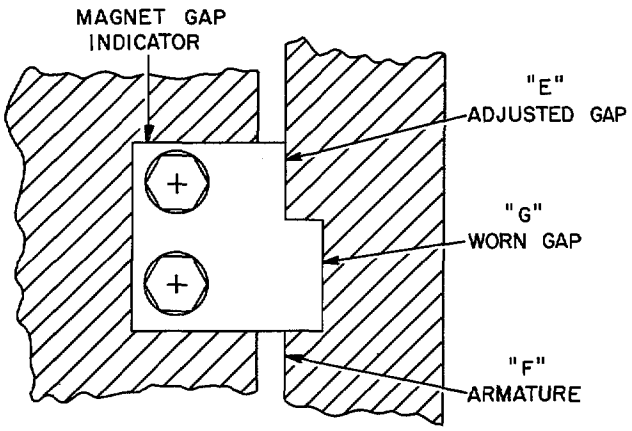


Fig. 4. Magnet gap alignment

CONNECTIONS

Shunt-wound brakes require the use of a series resistor which is separately mounted. The resistor is furnished with the brake when the brake is purchased separately. It is furnished as part of the control panel when the brake and panel are purchased together. Do not place a separate resistor in the brake circuit if a resistor is provided with a panel.

For use on systems of 115 to 250 volts, the resistor is supplied with taps for 115, 125, 230 and 250 volts. For higher voltages, the resistor furnished is designed to suit the voltage.

The brake nameplate shows the current rating of the coil and the ohmic value of the separately mounted resistor. Coil leads are brought to a terminal block under a conduit box at the magnet side of the brake.

Shunt-wound brakes will release at 80 percent of rated voltage, and will operate satisfactorily at 110 percent of rated voltage.

Connect series-wound brakes directly in series with the motor as indicated on the wiring diagram of the controller used. Series-wound brakes will release at 40 percent of rated current or less, and will set at 10 percent or less.

MAINTENANCE

LINING WEAR ADJUSTMENT

Check the magnet gap indicator (Fig. 4) for lining wear from time to time. As long as the armature edge "F" lies between "E" and "G", no adjustment is necessary. If the armature edge extends beyond edge "G", the magnet gap is too great and may result in slow release or no release.

To correct for lining wear, first loosen the equalizer bolt "C" (Fig. 3); then, turn the gap adjustment nut "D" clockwise until the armature edge "F" (Fig. 4) lines up with the magnet gap indicator edge "E". Retighten the equalizer bolt "C".

MANUAL RELEASE

To release the brake manually, tighten the release nut (4) Fig. 1 until the armature is pulled back against the core face. This will draw the brake shoes clear of the brake wheel. Restore the original setting for normal operation.

SHOE LINING REPLACEMENT

To remove worn linings, release the brake manually and remove the bolts and upper clamp holding the upper end of the brake shoes. Insert a screwdriver in the hole near the top of the lining which is located in the steel backing; then, slide the lining and backing assembly upward around the brake wheel:

As additional clearance is required between shoes and wheel when installing new linings, loosen bolt "C" (Fig. 3); then, turn the gap adjustment nut "D" to provide additional shoe-to-wheel clearance. After sliding the new linings into position and clamping them into place, readjust the operating gap by turning the gap adjustment nut until the armature edge lines up with the adjusted gap portion of the gap indicator. (See Fig. 4.) Then, retighten the equalizer bolt "C".

CORE-COIL ASSEMBLY REPLACEMENT

The coil is wound directly on the magnet core which in turn is bolted to the back plate over a thin non-magnetic shim. To remove the core and coil, it is first necessary to remove the conduit boxes and disconnect the coil leads from the terminal boards. Then, back-off on the two torque springs until all pressure is removed. Hook a cable over the upper lifting ears on the backplate. Now, remove all four bolts securing the backplate to the magnet housing. The coil, core, and backplate assembly can now be removed. Re-assemble in the reverse order, making certain that the non-magnetic shim is replaced between the core and backplate.

MAGNET GAP CLEANING

If shoe clearance around the wheel becomes reduced or the shoes drag, remove the ring-shaped dust shield mounted over the edge of the armature and check for any accumulation of dirt between armature and core. Blow out any accumulated dirt with dry, compressed air. If this is not possible, back-off

the armature sufficiently by means of the gap adjustor to insert a hacksaw blade or similar tool and scrape the core and armature clean. Readjust the gap to specified dimensions.

WHEEL REMOVAL

To remove the wheel, release the brake shoes from the wheel by tightening the manual release nut. (See Fig. 1.) First, back-off the nut (12) to relieve its spring pressure back against the yoke and then loosen the gap adjustor (1A). This will free the cross-bar from the yoke so it can be swung back out of the way. Now, loosen the four brake bolts so that the shoes are free to pivot. The wheel and motor shaft can be lifted vertically. As the wheel is moved upward, the shoes will pivot and allow the wheel to be lifted clear. After replacing the wheel, back-off the manual release, allowing the shoes to seat on the wheel. Then, swing the cross-bar back into position in the upper slot of the yoke. Snug-up the gap adjustor (1A). Then, retighten the nuts (12) until the length of the small spring on the upper tie bar is 2-1/2 inches long. Now, make the final setting of the gap adjustor to the required setting. Then, tighten all four shoe bolts.

LUBRICATION

As all moving pins operate in special graphite-impregnated bronze bearings and are sealed to prevent entry of dirt and grime, no lubrication is necessary.

RENEWAL PARTS

Order renewal parts from the nearest Apparatus Sales Office of the General Electric Company. Give the complete nameplate rating of the brake and describe in detail the part required.

INDUSTRY CONTROL DEPARTMENT

GENERAL ELECTRIC

SALEM, VA. 24153

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